How is it that the universe began in such a highly ordered configuration, setting things up so that for billions of years to follow everything could slowly evolve through steadily less ordered configurations toward higher and higher entropy? Don't lose sight of how remarkable this is. As we emphasized, from the standpoint of probability it is much more likely that the partially melted ice cubes you saw at 10:30 p.m. got there because a statistical fluke acted itself out in a glass of liquid water, than that they originated in the even less likely state of fully formed ice cubes. And what's true for ice cubes is true a gazillion times over for the whole universe. Probabilistically speaking, it is mind-bogglingly more likely that everything we now see in the universe arose from a rare but every-so-often-expectable statistical aberration away from total disorder, rather than having slowly evolved from the even more unlikely, the incredibly more ordered, the astoundingly low-entropy starting point required by the big bang.[19]

Yet, when we went with the odds and imagined that everything popped into existence by a statistical fluke, we found ourselves in a quagmire: that route called into question the laws of physics themselves. And so we are inclined to buck the bookies and go with a low-entropy big bang as the explanation for the arrow of time. The puzzle then is to explain how the universe began in such an unlikely, highly ordered configuration. *That* is the question to which the arrow of time points. It all comes down to cosmology.[20]

We will take up a detailed discussion of cosmology in Chapters 8 through 11, but notice first that our discussion of time suffers from a serious shortcoming: everything we've said has been based purely on classical physics. Let's now consider how quantum mechanics affects our understanding of time and our pursuit of its arrow.